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Empirical research

Derived rule-following and transformations of stimulus function in a children's game: An application of PEAK-E with children with developmental disabilities

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ABSTRACT

Derived relational responding in traditional match-to-sample tasks has been well documented in the behavioral scientific literature with children with developmental disabilities, but less is known regarding derived rule-following and corresponding transformations of stimulus function. The present study evaluated the efficacy of the sequential delivery of rules on children's mutual entailment of synonymous anatomical terms, as well as the transformation of stimulus function in the context of a common board game (i.e., *Twister*). The results of the study suggest that rule statements in the form of "*x* is the same as *y*" and "*y* is the same as *z*," where *x* was a known anatomical term and *y* and *z* were unknown synonymous anatomical terms, improved participant game performance when synonymous anatomical terms (*z*) were used. The results expand upon existing research by demonstrating derived rule-following and the transformation of stimulus function with a little researched population.

1. . Introduction

In the time span between 1997 and 2008, the prevalence of children diagnosed with developmental disabilities increased by 17.1%, or approximately 1.8 million children (Boyle et al., 2011). Estimates as of 2008 suggest that 15% of children have some form of developmental disability, which translates to approximately 1 out of every 6 children between the ages of 3- and 17-years of age. Children with developmental disabilities characteristically show deficits in communication and language, where the severity of such deficits are related to aggression (Chung, Jenner, Chamberlain, & Corbett, 1995) and other problem behaviors (Sigafoos, 2000) commonly displayed by developmentally disabled children. Skinner provided the first comprehensive naturalistic account of language from a behavior analytic perspective in his book *Verbal Behavior* (1957), which has been the foundation for an increasing body of empirical analyses supporting the use of behavioral technologies for promoting language development (Dymond, O'Hora, Whelan, & O'Donovan, 2006; Sautter & LeBlanc, 2006), especially in application with individuals with developmental disabilities (Dixon, Small, & Rosales, 2007). A limitation in the extant literature however is that applied studies have primarily focused on the most elementary forms of verbal behavior (Dymond et al., 2006). Empirical support for simple language skills such as the tact and mand have been researched almost to the point of redundancy (Dymond et al., 2006, p. 75), while research on developing complex language skills has not generated the same support. Several authors have suggested that Skinner's account has failed to produce empirical support in terms of complex language in part because his account is too broad so as to lend itself to empirical investigation through experimental manipulation (e.g., Barnes-Holmes, Barnes-Holmes, & Cullinan, 2000; Hayes, & Barnes-Holmes, 2004; Hayes, Blackledge, and Barnes-Holmes, 2001), which may also explain the relative success of contemporary theories of language, such as stimulus equivalence (Sidman & Tailby, 1982; Sidman, 1971) and Relational Frame Theory (Hayes et al., 2001).

Like Skinner's theory of verbal behavior, contemporary theories have also produced a growing body of empirical support in terms of language development (Dymond, May, Munnelly, & Hoon, 2010). Unlike Skinner's theory of verbal behavior, a majority of the empirical research supporting contemporary approaches to language development have been conducted in application with typically developing populations (Dymond et al., 2010). The scope of contemporary theories in terms of providing an analysis of complex language is made evident through application with typically developing participants; however, the application of these theories with participants with developmental disabilities requires further empirical investigation, especially as procedures tailored specifically for use with this population continue to be developed (see Rehfeldt & Barnes-Holmes, 2009). One area of research that may have especial utility in application with such populations is rule-governed behavior (see Hayes, 1989) and derived rule following. Contemporary theories of verbal behavior and rule-governance deviate from traditional Skinnerian formulations (see *Verbal Behavior*, Skinner, 1957) primarily in terms of the inclusion of listener behavior in the analysis (Hayes & Hayes, 1989). By considering the behavior of the listener as something other than merely a mediator of contingencies, such issues as meaning, understanding, and

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reference can be handled in a manner not afforded previously (Parrott, 1984).

In considering listener behavior, rule-governed behavior involves the presentation of a verbal stimulus at one time (e.g., *if the light is red, then do not drive through the light*), where the listener responds in terms of the verbal stimulus at another time (e.g., stopping at a red light when driving). According to Torneke, Luciano, and Salas (2008), the verbal stimulus of the rule serves to transform the function of stimuli that are in the environment. The red light which evoked no specific set of responses prior to the rule now evokes stopping behavior in the absence of direct exposure to the contingencies that would otherwise govern stopping (i.e., crashing into oncoming vehicles). Rule governance is a sophisticated phenomenon that is possible due to the symbolic nature of human language (Torneke et al., 2008). In the example above, the words *light*, *red*, and *drive*, bear no formal similarity to the physical stimulus light, the physical property red, nor the physical action of stopping, and yet these words have meaning to the speaker and are understood by the listener. Words, such as those above, acquire meaning through the derivational processes of mutual entailment, combinatorial entailment, and the transformation of stimulus function (Hayes et al., 2001). Mutual entailment involves the bidirectional relationship between stimuli. For example, if taught that the word *red* is related to a visual stimulus red, an individual will also relate a visual stimulus red to the word *red*. Combinatorial entailment extends upon mutual entailment, and involves the bidirectional relationship between stimuli based on their relationship to other stimuli. To continue with the example of red, if taught that *red* means the same as *stop*, then an individual will also relate the visual stimulus of red with the word *stop*. A transformation of stimulus function, then, occurs when the function of a previously neutral or otherwise conditioned stimulus is transformed in terms of another stimulus. For example, if a driver is likely to stop the car when told to *stop*, then the driver will now likely stop the car when they see a red light or when a passenger anxiously states *the light is red*. Again, the word *red* nor the red light bear formal similarity to the word *stop* nor the action of stopping. Provided a sufficient history of relating words with physical objects and events, however, individuals are able to demonstrate rule-governed behavior of the sort discussed by Hayes et al. (1989), Torneke et al. (2008).

Derived rule-following then occurs when participants respond to a rule, like the one presented above, when the totality of the rule was never directly stated. In line with the above example, consider a scenario where an individual is told that *light means the same as bulb*. Given a previous rule of *if the light is red, then do not drive through the light*, the individual would likely stop the car when a passenger anxiously states *the bulb is red*. Although this may appear as a product of simple stimulus generalization, as *the light is red* is formally similar to *the bulb is red*; an anxious passenger stating *the doll is red* is not likely to result in the individual stopping the car, but may instead result in a trip to the store where a doll may be purchased. Derived rule following therefore serves as an acute example of the symbolic nature of language and how language affects overt behavior, and may provide further insight into say-do correspondence between rules and behavior (e.g., Baer, Detrich, & Weninger, 1988; Luciano, Herruzo, & Barnes-Holmes, 2011). The most basic form of derived rule following may occur in the case of synonyms, which theoretically necessitates only a frame of coordination or equivalence between one arbitrary stimulus and another, as well as an observed transformation of stimulus function in the context of the rule. Tarbox, Zuckerman, Bishop, Olive, and O'Hora (2011) demonstrated that individuals with disabilities could be taught to follow rules through multiple exemplar training, but further research is needed to show how, when children are already able to demonstrate rule-following, how the development of frames of coordination between known rules and synonymous terms interact or transform behavior.

Therefore, the purpose of the present study was to evaluate the degree to which the delivery of rules resulted in the development of frames of coordination, as well as a transformation of stimulus function in the context of rule-following during a children's game (i.e., *Twister*). In the study, children were taught synonyms for body parts (e.g., hands and feet), and testing was conducted to determine the degree to which participants were able to use the correct body part when provided only the synonym and an action in a game of *Twister. Twister* was selected as the task because it was likely a familiar game for the participants in the study, and the game did not provide any textual or otherwise visual prompts that would suggest the relationship between the synonymous words and the corresponding responses were anything but arbitrary. Training, in support of contemporary theories that emphasize the behavior of the listener, involved only listener behavior on the part of the participant and no reinforcement for correct responding. The training procedures in the study were taken directly from the *Promoting the Emergence of Advanced Knowledge Equivalence Module* (PEAK-E; Dixon, 2015) to provide standardized procedures that can be easily replicated both in future studies and in clinical practice. PEAK-E is comprised of an assessment of participants' relational abilities and a collection of 184 programs designed to progress participants' relational skills based on their results on the assessment. The procedures were taken from one of the 184 programs, *12M: Equivalence – Body Parts Game,* which adopts a linear approach to rule delivery (LS; A-B, B-C). Other approaches to relational training, such as one-to-many (OTM; A-B, A-C) and many-to-one (MTO; B-A, C-A), have both demonstrated faster acquisition and greater maintenance of entailed relations compared to an LS approach (Arntzen & Holth, 1997), suggesting that relating stimuli in a linear fashion may be more difficult. Because derived rule following is one of the more advanced skills targeted in the PEAK-E curriculum, the participants were required to relate linearly. A secondary purpose of the present study was therefore to evaluate the efficacy of the PEAK-E curriculum in application with children with developmental disabilities.

2. Method

2.1. Participants and setting

Three male students with developmental disabilities participated in the study. John (age 9) was diagnosed with an intellectual disability, Paul (age 7) was diagnosed with an emotional and learning disability, and George (age 8) was undergoing evaluation for an autism diagnosis. Participants received 1- to 2-hours of behavior analytic programming daily while attending a special education classroom at a Midwest American elementary school. Each of the participants were receiving special education services per their individualized education plans. Prior to the study, each participant demonstrated the ability to read and write basic words and engage in conversation. Participants were selected for inclusion in the study due to failure to name or receptively identify synonyms for known anatomical terms (e.g., *cranium* in place of *head*). This was assessed by showing the participants pictures of a head, foot, and hand, and asking them to provide three names for each of the picture stimuli. The participants were then provided the pictures in a three picture stimulus array, and were asked to receptively label the pictures using B and C stimuli (see below). Because John was able to receptively label one of the picture cards when provided the synonymous anatomical term, a different anatomical term was used in the study to ensure that a prior relational history did not affect the results. No other students at the school were tested for inclusion in the study. The experiment took place in the elementary school gym while no other students were present. Sessions were conducted by the authors of the study, and each session lasted approximately 30-minutes, where either 1- or 2- sessions were conducted within a single day. One trial block was conducted within each session, where there were 5 trials within each block.

3. Materials

Materials used in the present study were those outlined in PEAK-E program *12M: Equivalence – Body Parts Game* from the PEAK-E curriculum (Dixon, 2015). This program provided the experimenters with a task analysis of how to conduct the training, as well as how to test for derived responding and transformations of stimulus function. The materials used in the present study included a large plastic *Twister* mat and a cardboard *Twister* spinner. Images of a standard *Twister* mat and spinner are displayed in Fig. 1. The *Twister* mat included rows of five circles, where each row exclusively contained red, blue, yellow or green circles. The *Twister* spinner had a spinning arrow in the center, around were circles corresponded with the colors on the *Twister* mat. On the outside of the spinner were paper inserts with text and picture stimuli denoting anatomical body parts (i.e., hand, foot, and head). Anatomical words for this study included *hand, foot,* and *head* (A stimuli) and synonyms for these words that were unknown to the participants prior to the study (B and C stimuli). The words used for each participant are displayed in Table 1.

4. Dependent variable and inter-observer agreement

The dependent variable in the study was the percentage of correct responses emitted within a trial block, where each block consisted of 5-trials. Correct responses occurred when participants touched the correct circle color with the correct body part, as specified by the experimenter. For example, if the experimenter said "Put your *foot* on *yellow*," then a correct response occurred if when a participant contacted the yellow circle with their foot prior to contacting the yellow circle with any other targeted body part, within 10-s of the delivery of the discriminative stimulus. The number of correct responses was divided by 5, and multiplied by 100, to determine the percentage of correct responding. An independent observer scored 34% of the trials during baseline and intervention phases. Agreement occurred when both observers scored the same response on a given trial. Independent observer agreement (IOA) was scored by dividing the number of agreements by the total number of agreements plus disagreements and multiplying by 100 to obtain a percentage. Using this method IOA was calculated at 99%.

5. Design and procedure

An ABCDEFG multiple baseline design across subjects was used to evaluate the efficacy of the relational training on emergent transformations of stimulus function. Across all phases, participants did not progress to the next phase until a decreasing trend or steady responding was observed in the previous phase. Steady-state responding was defined as participant scores within 20% of the previous data point. The A phase consisted of a pre-training test probe to evaluate if participants were able to play *Twister* using known anatomical words (A). Following the A phase, a B baseline phase was conducted to evaluate if participants were able to play *Twister* using unknown synonymous anatomical words (C). The C, D, and E phases were mixed A-B and B-C training phases, where each phase involved training of a single class of anatomical terms (e.g., A1–B1; B1–C1). Training was conducted for *hand* in the C phase, *foot* in the D phase, and *head* in the E phase. The F phase was a maintenance phase and the G phase was an instructor reversal phase.

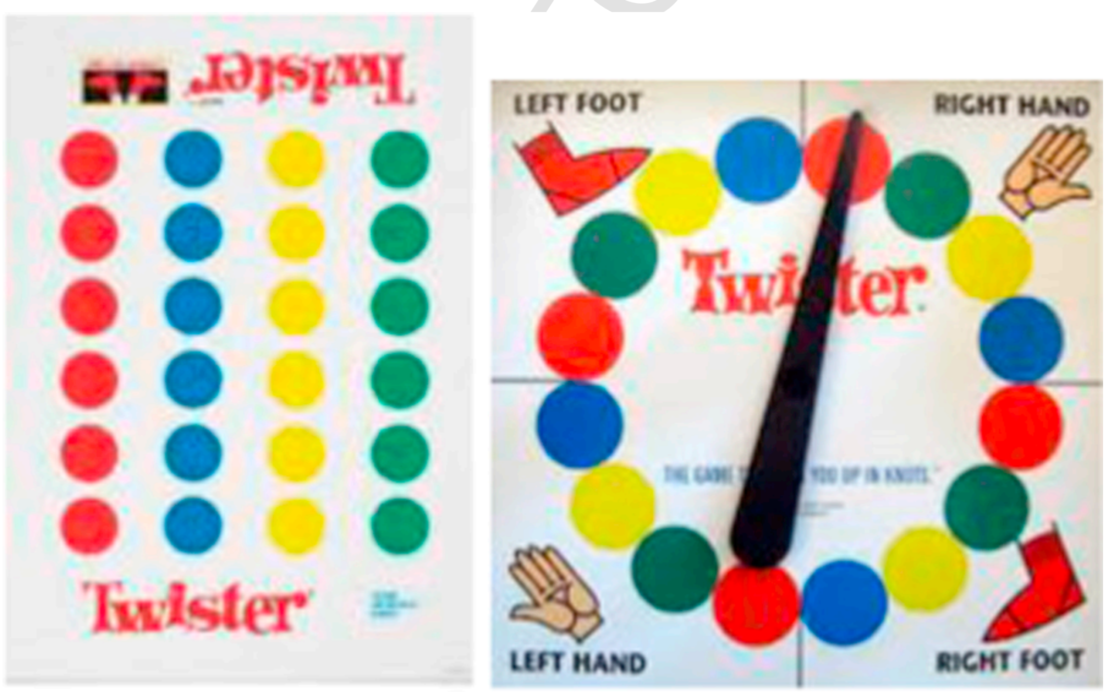

Fig. 1. Pictures of a standard *Twister* mat and *Twister* spinner. The anatomical terms on the standard spinner (e.g., left foot, right hand) were covered with paper inserts including the text and picture stimuli for *foot, head, hand,* and *re-spin*.

Table 1.
Vocal stimulus classes in the present study.

Class/stimulus	A	B	C
1			
John	Hand	Palm	Mitt
Paul	Hand	Palm	Phalanges
George	Hand	Palm	Phalanges
2			
John	Foot	Tootsies	Sole
Paul	Foot	Tootsies	Sole
George	Foot	Tootsies	Sole
3			
John	Head	Noggin	Dome
Paul	Head	Noggin	Cranium
George	Head	Noggin	Cranium

5.1. Pre-training test and baseline phases

In the pre-training test phase, participants were required to play *Twister* using known anatomical terms (A), including: *hand, foot,* and *head*. The game involved the experimenter spinning the *Twister* spinner such that it landed on a color and anatomical term. The spinner was not visible to the participants. The experimenter then delivered the multiple discriminative stimulus "put your *x* on *y,"* where *x* was an anatomical word (A) and *y* was the color. All participants responded with 100% accuracy during this initial phase. The pre-training test phase was conducted to ensure that each participant was able to correctly respond to the multiple discriminate stimulus in a game of *Twister* when known anatomical terms were used. No correction procedures were required as each participant demonstrated mastery in this phase. The second phase was a baseline phase where participants were required to play *Twister* using unknown synonymous anatomical words (C). Again, the game involved the experimenter spinning the *Twister* spinner such that it landed on a color. The experimenter then delivered the multiple discriminative stimulus "put your *x* on *y,"* where *x* was an anatomical word (C) and *y* was the color. The baseline phase was conducted to determine if participants were able to correctly respond to the multiple discriminative stimulus when unknown synonymous anatomical terms were used.

5.2. A-B and B-C training phases

Following the pre-training test and baseline phases, A-B and B-C training was conducted where participants were given rule statements in the form of "*x is the same as y,"* where *x* was an anatomical word and *y* was a synonymous anatomical word. Following the A-B and B-C rule statement, the experimenter probed for mutual entailment by asking, "*what is the same as y? "* No feedback was provided to the participants. In each training phase, A-B training was conducted immediately prior to B-C training. In the C training phase, the experimenter told the participants that *hand* is the same as *palm* (A-B). Immediately following A-B training, B-C training was conducted whereby participants were told that a *palm* was the same as an unknown anatomical word (B-C). *Phalanges* were used as the synonymous anatomical word for Paul and George and *mitt* was used for John. Following the delivery of these two rule statements, the C phase was conducted in the same manner as in baseline. The D phase began following the C phase. In the D training phase, participants were told that *foot* is the same as *tootsies* (A-B) and that *tootsies* are the same as *sole* (B-C). Again, the D phase was conducted in the same manner as in baseline following the delivery of the two rule statements. Following the D phase, training began for the E phase. In the E training phase, participants were provided the rules that *head* is the same as *noggin* (A-B), and that *noggin* was the same as an unknown anatomical word (B-C).

5.3. Follow up and instructor reversal phases

Two- and three- weeks following the conclusion of the E training phase, a follow-up phase and an instructor reversal phase were conducted. The conditions of the follow-up phase were identical to the baseline phase, and this phase was conducted to determine the degree to which the results observed in the training phases were maintained in the absence of training. Two follow up probes were conducted for John, and one probe was conducted for the remaining participants. For John, the follow-up probes were conducted on the same day. One-week following the conclusion of the follow-up phase, an instructor reversal phase was conducted for each of the participants. In the instructor reversal phase, the instructor played *Twister* and the participant served as the experimenter. This phase was conducted to switch the speaker-listener roles. In this phase, the participant was required to spin the wheel and say "put your x on y," where *x* was an anatomical term and *y* was the color that the spinner landed on. If the participant provided an A stimulus for *x*, then the instructor did not move their body part to the corresponding circle, but rather prompted the response by asking "What else could you call that? " A response was considered correct if the participant emitted the corresponding C stimulus. Throughout all phases in the present study, no reinforcement was provided for correct responding. Intermittent reinforcement was delivered non-contingently in the form of specific praise for participating as well as breaks or edibles were available upon request.

6. Results

In the pre-training test phase, each of the participants demonstrated 100% correct responding during a regular game of *Twister,* when known anatomical words were used. These results suggest that all participants were familiar with the game rules, and these data were not depicted in the below pictures to maintain consistency in the representation of the obtained data. The results of the present study also suggested that each of the participants were able to demonstrate mutual entailment responses with 100% accuracy following the delivery of A-B and B-C rule statements. The results of participants' derived rule following are shown in Figs. 2 and 3. Fig. 2 shows the overall percentage of correct responding within each 5-trial block across each of the phases. In the baseline phase, John, Paul, and George had mean percentage correct scores of 26.6, 5, and 20 respectively. If participants responded by guessing which of the three body parts to use, an average score of 33.3 would be expected by chance alone. Following A-B and B-C training in the *hand training* phase, mean percentage correct scored increased to 60, 53.3, and 33.3 for John, Paul, and George. In each phase, percentage of non-overlapping data (PND) values were evaluated relative to the prior phase, in order to provide a metric for evaluating increases in correct responding within each phase. PND was calculated by determining the percentage of data points that were greater than the highest data point in the previous phase. Relative to baseline, John's PND was 75-%, Paul's PND was 100-%, and George's PND was 0-%. Following training in the *foot training* phase, each of the participants mean percentage correct increased relative to the previous phase. John's score and Paul's score both increased to 73.3, and George's score increased to 80. The PND for the participants were 0, 66.7, and 100, respectively. In the final training phase, all of the participants demonstrated 100% correct responding; resulting in PNDs of 100 for each participant. In the follow-up phase, scores for Paul and George remained at 100%, and scores for John dropped to 90%. Similarly, in the instructor reversal

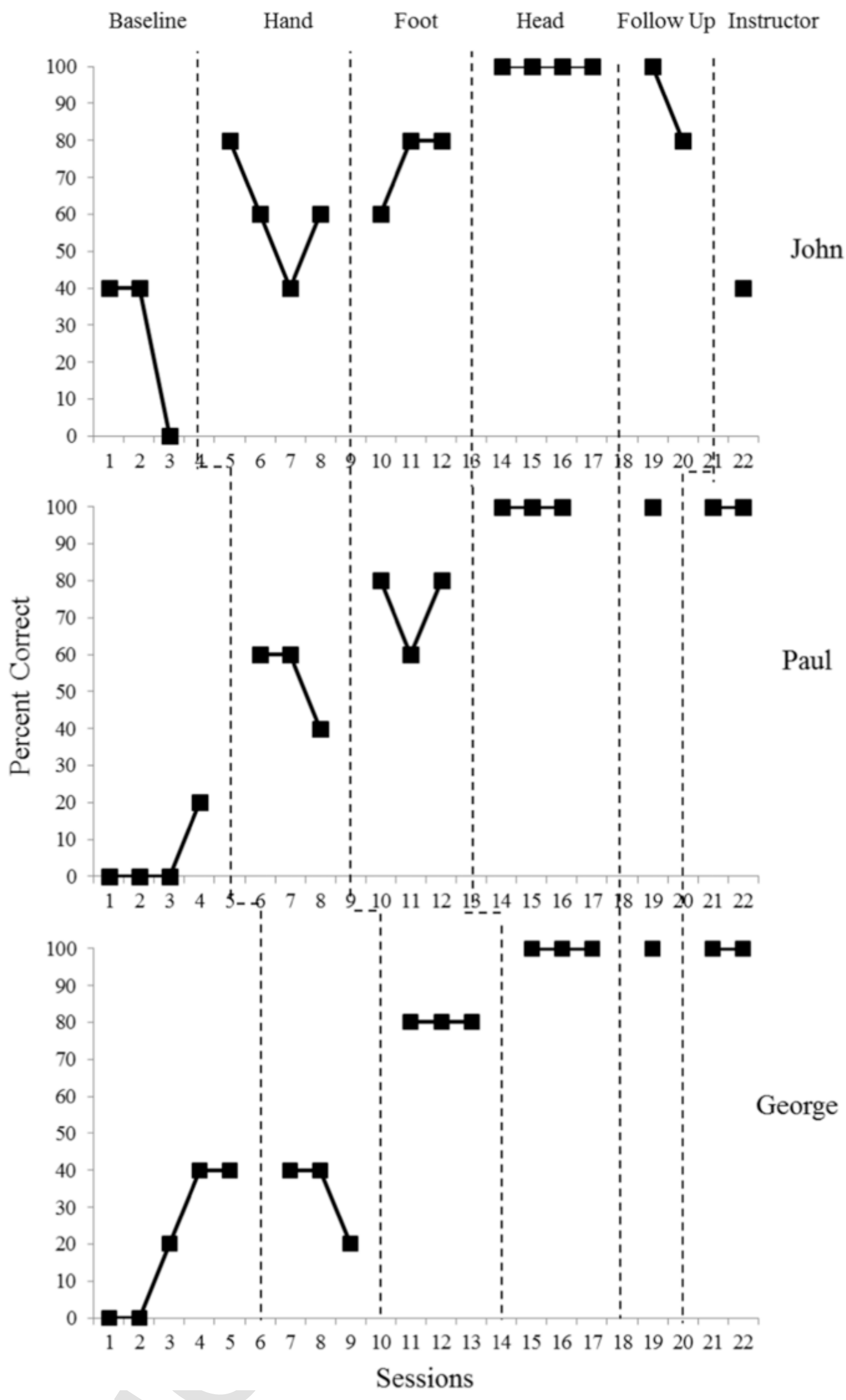

Fig. 2. Percentage of correct responses emitted while playing Twister across phases.

phase, Paul and George demonstrated 100% correct responding, whereas John demonstrated 40% correct responding.

Fig. 3 shows the percentage of correct responding for each stimulus class separately (i.e., *hand, foot, head*). This was done because the data in the above figure demonstrated the overall percentage of correct responding rather than the correct responding within each body part class. The data from Fig. 3 show that immediately following training, the stimulus class that was trained immediately increased to 100% correct responding. In baseline, no anatomical terms resulted in 100% correct responding, whereas Mitt/ Phalanges resulted in 100% correct responding in phase C; *Mitt/Phalanges* and *Sole* resulted in 100% correct responding in phase D, and *Mitt/Phalanges, Sole,* and *Dome/Cranium* resulted in 100% correct responding in phase E.

7. Discussion

The results of the present study expand upon the existing literature by demonstrating derived rule following in three children with devel-

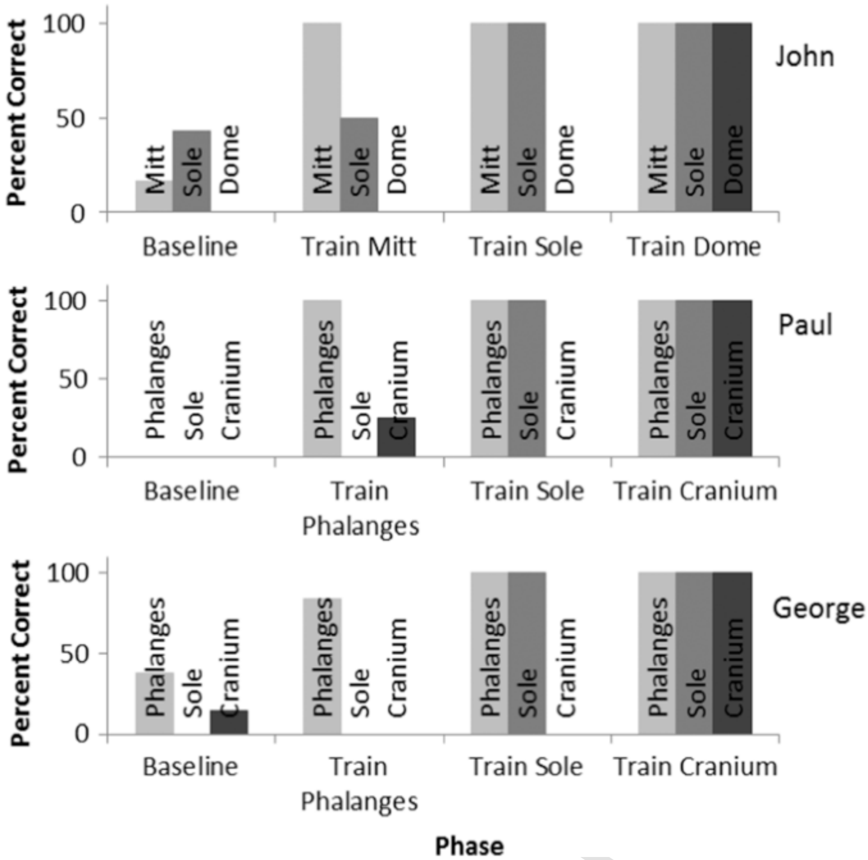

Fig. 3. Percentage of correct responses emitted for each body part across baseline and introduction of each rule statement for each of the three participants.

opmental disabilities, where only listener responding was required in the development of frames of coordination and corresponding transformations of stimulus function. During a regular game of *Twister*, all of the participants were able to demonstrate correct responding by moving the specified body part to a specific color when vocally provided the anatomical word and the color in a rule statement. The results of the baseline phase suggested that the participants were not able to demonstrate the same correct responding when synonymous anatomical terms were used in place of the known anatomical terms. The results of the relational training, as shown in Fig. 3 show that the mutually entailed B-A and C-B responses were only demonstrated by each of the participants following the corresponding rule delivery. This suggests that the procedures were efficacious in teaching the mutually entailed relations in a stepwise manner as intended in the study. Following the delivery of a rule stating that the known anatomical words were the same as synonymous anatomical words, and that the synonymous anatomical words were the same as other synonymous anatomical words, all of the participants were able to respond correctly when provided the synonymous words in place of the known anatomical words. This result demonstrates that not only were participants able to derive mutual entailment between the known word and their synonyms, participants were also able to demonstrate a resulting transformation of function through the derivation of a new rule (for review of transformations of stimulus functions, see Dymond & Rehfeldt, 2000, Hayes et al., 2001). The results of the follow-up condition suggest that learning was largely maintained following 2-weeks, and that two of the three participants were not only able to demonstrate listener behavior as the player, but also bidirectional speaker behavior as the instructor.

There were several limitations that restrict the inferences that can be generated from the present study. A first limitation was that only three stimulus classes were taught to each participant. The number of relations was restricted to three to demonstrate tight experimental control over the taught relations; however, more relations would have led to a more discernable rate of learning and acquisition resulting in a more robust analysis. The addition of more relations would have provided more topographies of the response, although the present data were sufficient to demonstrate the function and transformation of function from one controlling variable to another. A second limitation was that combinatorial entailment was not directly tested. Testing of the combinatorially entailed A-C and C-A relations were not conducted to avoid testing effects exerting influence on participant behavior during the transformation of function probes. Conceptually, the transformation of stimulus function would not occur in the absence of combinatorial entailment, but including this analysis could add to the certainty of the processes underlying the obtained results. A third limitation was that, by only training a single relational class (e.g., A1-B1 and B1-C1) in each phase, sequence affects may have been introduced as part of the design. This is evidenced by the stepwise increases in correct responding within each phase as more relational classes were taught. A fourth limitation was that the training effects were not well maintained during the follow-up phase for John after a relatively short period of time had elapsed. This outcome, although not a central focus of the present study, does suggest that further evaluation is needed to determine approaches that not only promote the transformation of stimulus function, but also the maintenance of this transformation. A final limitation was that the procedures were implemented by trained graduate students who had previous training in both the conceptual underpinnings and implementation of the procedures in the PEAK-E curriculum. Using trained graduate students provided increased confidence in the fidelity of program implementation, but may have decrease the external validity of the obtained data as PEAK-E is intended to be implemented by both trained professionals as well as implementers with minimal training. Future research may address the limitations of the present study, as well as expand upon the presented data. Future research may compare the instructional process of delivering rule statements, like those in the present study, with match-to-sample procedures that are common in stimulus equivalence training programs (e.g., Ramirez & Rehfeldt, 2009). The pervasiveness of match-to-sample procedures is likely a product of traditional approaches in the application of behavioral techniques with lower-functioning participants, however the delivery of rule statements may be more common in regular instruction and more efficient in delivery. The results of the present study suggest the delivery of rule statements may be sufficient for some participants, but it is likely the case that not all participants will possess the requisite language skills to learn from such an instructional approach. Therefore, future research may evaluate what skills are predictive of success in this arrangement so as to optimize relational training of synonyms as well as other more advanced relations.

In summary, Skinner (1957) discussed several complex forms of verbal operant behavior that have not generated the same empirical support as the more elementary forms of verbal operant behavior pervasive in the behavior analytic literature in application with children with developmental disabilities (Dymond et al., 2006). Contemporary theories of complex human behavior, such as Relational Frame Theory, provide a model that may further explain the nuances and complexities of human language, but further research is needed in when such approaches are used with developmentally disabled participants (Ruiz & Montoya, 2014). The results of the present study demonstrated the development of frames of coordination through rule-statements, as well as derived rule following, resulting in the transformation of stimulus functions in a game of *Twister*. In addition, the methods were taken from the PEAK-E curriculum, which may improve the replicability of the obtained results both in clinical practice and in research.

Disclosure

The first author receives small royalties from sales of the PEAK curriculum.

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